GATE Counseling, Admission Process & PSU Recruitment

The Complete Student Guide for ECE/EE Engineers

GATE 2026 Edition

Continuously Updated • Community-Sourced

Compiled by **PrepFusion**

Special thanks to Srijan Dwivedi, Palak Agarwal,
Sangrag, Nilimay, Ravi, Deepto & the PrepFusion Community

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This document is dynamic and will be continuously updated as new information is received from the PrepFusion student community.

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Part I

M.Tech Admissions

Chapter 1

Counseling & Admission Process

Pro-Tip: Most Frequently Asked

This chapter addresses the most common questions about COAP (for IITs & IISc) and CCMT (for NITs). Read it before applying to ensure you don't miss critical deadlines or steps.

1.1 Overview of the Two Portals

After the GATE result is declared, admissions to postgraduate programs are channeled through two centralized portals depending on the institution:

Portal	Institutions Covered	Opens Approximately
COAP (Common Offer Acceptance Portal)	IITs, IISc	Last week of March
CCMT (Centralized Counseling for M.Tech)	NITs, IIEST Shibpur, select IIITs & CFTIs	Last week of May

M.Tech classes typically begin at the **end of July or August**.

1.2 The COAP Process (IITs & IISc)

1.2.1 Step-by-Step Workflow

Step 1. **Apply on institute websites.** Visit each desired IIT/IISc portal separately, select your preferred branch, and submit your application along with your GATE score.

Step 2. **Register on the COAP portal.** Create a single COAP account—all offer communications will appear here regardless of how many institutes you have applied to.

Step 3. **Attend interviews (where required).** Some institutes conduct interviews before shortlisting. If shortlisted, you will receive instructions via email from the respective institute. **Physical attendance is mandatory for interviews.**

Step 4. **Check offers each round.** Accepted and pending offers become visible after each COAP round. You may **hold** an offer for up to **2 rounds** while waiting for a better offer.

Step 5. **Accept, hold, or decline.** From your pool of concurrent offers, select the most preferred one. Accepting an offer removes all other offers automatically.

Step 6. **Freeze your seat.** Once you are satisfied, freeze the offer. The institute will then send admission instructions and a fee payment link.

Pro-Tip: Don't lose your hold window

An offer can be held for a maximum of **2 consecutive rounds**. If you neither accept nor decline within this window, it is automatically withdrawn. Track the round schedule carefully.

1.2.2 Interviews at IITs: Which Institutes Require Them?

Not all IITs require interviews for M.Tech admissions. The table below summarizes the current policy (verify on official institute websites each year):

Institute	Interview for M.Tech?	Notes
IISc Bangalore	**Yes (all branches)**	Interview mandatory for every M.Tech & M.Tech Research branch.
IIT Delhi	Partial	IEC branch requires interview; check other branches officially.
IIT Kharagpur	Partial	EE department has interview; ECE department generally does not.
IIT Bombay	**No (M.Tech)**	No interview for 2-year M.Tech VLSI branches; MS (3-yr) requires interview.
IIT Madras	**No (M.Tech)**	Similar to IITB; ICS/VLSI M.Tech has no interview. MS and TI-MS require one.
Other IITs	Generally No	MS/M.Tech Research programs at *all* IITs require interviews.

1.3 The CCMT Process (NITs & CFTIs)

The CCMT process operates similarly to COAP but serves NITs, IIEST Shibpur, select IIITs, and Centrally Funded Technical Institutes (CFTIs).

- Register on the CCMT portal after GATE results.
- Fill in your choice of institute and branch in order of preference.
- Seat allotments happen in multiple rounds; participate in document verification and fee payment to confirm your seat.
- Check the official CCMT portal for round-wise cutoffs: `https://ccmt.admissions.nic.in`

Pro-Tip: NIT Cutoff Reference

A community-sourced NIT cutoff spreadsheet (CCMT 2025) is available at: `https://docs.google.com/spreadsheets/d/1G0KjJN9MNMiwWDRDf4OrzgEBYivGnn_lT0G7jK57ikk`

1.4 CFTI Mode Admissions

Eligibility Alert: CFTI Mode Eligibility

IISc and IIT Kanpur (and select other IITs) offer admissions via the **CFTI mode** for certain branches (e.g., Semiconductor Technology at IISc). Applicants are typically required to:

- ✓ Hold a degree from a CFTI institution.
- ✓ Have a CGPA of **8.0 or above**.
- ✓ Clear an interview (same panel as the GATE-based MS program).

Note on IIT Kanpur VLSI: It has been observed historically that all general category seats in the VLSI program are filled by IIT B.Tech graduates, leaving very limited seats for external GATE candidates. Verify the latest data before applying.

1.5 Documents Required

Each institute specifies its exact document list in the admission instructions. The following are universally required across most IITs and NITs:

- ✓ Class 10 (Secondary) mark sheet & certificate
- ✓ Class 12 (Higher Secondary) mark sheet & certificate
- ✓ All B.Tech/B.E. semester grade cards
- ✓ B.Tech/B.E. degree certificate (or provisional)
- ✓ GATE scorecard (current year)
- ✓ Proof of nationality (Aadhaar / Passport)
- ✓ Category certificate (SC/ST/OBC-NCL/EWS, as applicable)
- ✓ Recent passport-size photographs

1.6 Key Dates & Timeline

Important Date: GATE 2026 Counseling Timeline

Event	Approximate Date
GATE 2026 Result	March 2026
COAP Portal Opens (IITs & IISc)	Last week of March 2026
COAP Counseling Rounds	April – June 2026
CCMT Portal Opens (NITs & CFTIs)	Last week of May 2026
M.Tech Classes Begin	July / August 2026

1.7 Frequently Asked Questions

Q: If I clear the cutoff for colleges X and Y and apply to both, can I choose between them?

A: Yes. In COAP, you can receive multiple offers from various colleges in any round. You are free to choose your preferred branch and college from the pool of offers. However, once you **accept** one offer, all other offers disappear. Choose wisely—accept only when you are certain.

Q: How will I know if I am eligible for an interview at an institution?

A: If you are shortlisted, the institution will send instructions directly to your registered email address. Apply broadly to all institutes of interest—the shortlisting and communication happen automatically.

Q: Is physical travel required for IIT interviews?

A: **Yes.** As of current practice, all IIT and IISc interviews require the candidate to be physically present at the institute. Online modes have not been standardized across all institutes.

Chapter 2

College & Branch Selection

2.1 Top Institutes for VLSI / ECE

Tier	Institutes	Key Strengths
Tier 1	**IISc**, **IIT Bombay**, **IIT Madras**, **IIT Delhi**, **IIT Kanpur**	Best industry connections, research output, placement records, and faculty quality.
Tier 2	**IIT Kharagpur**, **IIT Roorkee**, **IIT Hyderabad**	Strong placements; IITH stands out among emerging IITs for placement, infrastructure, and teaching.
Tier 3	Remaining IITs	Vary significantly; check recent placement data per branch.

2.2 Best IIT for Analog Roles

Pro-Tip: Analog Domain Recommendation

IIT Madras is the premier choice for analog roles in placements and research. It has dedicated labs in RF, mixed-signal, and analog IC design, strong faculty with industry collaborations, and a track record of placing students in top semiconductor companies.

2.3 Top Courses at IISc by Placement

Institute	Top Branches (Placement-wise)
IISc Bangalore	CSE, AI, CDS, MVLSI, ESE, ECE, Signal Processing
IITs (general)	CSE, Data Science, VLSI, ICS, ESE

2.4 Pure VLSI vs. Related Fields

Q: Which field is better: pure VLSI or related fields like Electronic System Design (ESD) or Solid-State Devices?

A: Pure VLSI tends to be stronger in terms of campus placements. ESD/Signal Processing branches also have good placements at the top institutes. However, choice should ultimately be driven by **individual interest** and long-term career goals. Forced branch selection leads to poor academic performance.

Q: Should I choose Signal Processing at older IITs or VLSI at newer ones?

A: This depends on your personal interest. If you are passionate about Signal Processing, a strong older IIT is the better environment despite the branch. Placement data at newer IITs in VLSI has been improving but the faculty and research ecosystem of older IITs generally remains superior. Evaluate the specific institute's recent placement records.

2.5 NIT Trichy (Communication) vs. NIT Calicut (VLSI)

Both are strong institutes. Consider the following:

Factor	NIT Trichy – Communication	NIT Calicut – VLSI
Domain	Signal Processing, Wireless, Systems	Semiconductor design, CMOS, FPGA
Industry Demand	Telecom, DSP companies	Semiconductor, EDA, VLSI companies (higher demand)
Overall Placement	Excellent (top NIT)	Very good (strong regional industry connections)
Recommendation	Choose based on your interest in the domain; both are excellent choices.	

Chapter 3

GATE Score & Cutoff Analysis

3.1 IIT / IISc Cutoffs for VLSI Branches

Institute	Branch	UR Cutoff (Score)	Notes
IISc Bangalore	MVLSI	900+	Typically remains above 900 for UR in early rounds.
IISc Bangalore	ESE	850+	Slightly lower than MVLSI.
IIT Bombay	ICS	900+	Highly competitive; falls slightly in later rounds.
IIT Bombay	ESE	850+	
IIT Madras	ICS	850+	
IIT Delhi	IEC	720+ – 687	General cutoff starts ~720; drops to ~687 in final rounds.
IIT Hyderabad	VLSI	~650 (late rounds)	Significantly lower in final rounds.

Pro-Tip: Top-5 IITs + IISc General Rule

For the top-5 IITs and IISc, an approximate cutoff of **720+ GATE score** applies for General category candidates during the first 6 rounds. Cutoffs drop considerably in the final rounds.

3.2 Official Cutoff References

For official and updated cutoff data, refer to:

- **IISc:** https://iisc.ac.in/admissions/cut-off-report-and-gate-score-of-previous-
- **IIT Madras:** https://mtechadm.iitm.ac.in/pdfs/M.Tech.2024_Cutoff.pdf
- **IIT Kanpur (2024):** https://iitk.ac.in/ee/data/admissions/2024-25/Shorlist_Criteria-01-07-24.pdf
- **IIT Kanpur (2023):** https://www.iitk.ac.in/ee/data/admissions/2023-24/6_Shorlist_Criteria-26-06-23.pdf
- **NITs (CCMT 2025 Community Sheet):** https://docs.google.com/spreadsheets/d/1G0KjJN9MNMiwWDRDf4OrzgEBYivGnn_lT0G7jK57ikk

3.3 PSU Cutoff Reference

PSU	Category	Approximate Cutoff
NTPC	General	Rank $\lesssim$150 safe for top PSUs
NTPC	OBC	Round 2: approx. rank 1118
CIL	OBC	Round 3 availability at rank $\sim$1300–1400
NPCIL	OBC	563 GATE score (2024 cutoff)

3.4 Low Score Options (40 marks / $\sim$560 score)

Important Date: Opportunities at $\sim$40 Marks in GATE ECE

- ✓ **PhD at IITs/IISc:** Interview cutoffs for PhD are very low; admission is entirely based on the interview performance.
- ✓ **NIT VLSI branches via CCMT:** Check round-wise cutoffs on the CCMT portal; institutes like NIT Kurukshetra have good industry connections.
- ✓ **BITS Pilani HD program:** A strong option if fees are not a constraint (see Chapter 8).
- ✓ **NIT Kurukshetra example:** Proven track record of students securing internships and FTE offers despite being a newer-tier NIT.

Chapter 4

Financial & Loan-Related Queries

4.1 Fee Waivers & Scholarships

Eligibility Alert: EWS / Low-Income Students

EWS (Economically Weaker Section) students receive **almost full tuition fee waivers** at most IITs. Additionally, scholarships are available that can effectively sponsor the entire 2-year M.Tech course. Apply for EWS certificate well in advance of counseling.

4.2 Admission Fee Management

Q: Do all IITs require a ₹1000 admission fee? How can such expenses be managed?

A: Most IITs charge a non-refundable application fee in the ₹500–1500 range. When applying to multiple institutes, these costs add up. Strategies:

- ✓ Apply selectively—choose institutes where your GATE score is competitive.
- ✓ Note that the application fee is separate from the seat acceptance fee (which is refundable upon seat surrender in most cases).
- ✓ Education loans from nationalized banks can be availed before final admission confirmation in many cases; consult your bank branch directly.

4.3 Self-Sponsored Courses

Several IITs and NITs offer **self-sponsored / industry-sponsored** M.Tech seats with **significantly higher fee structures** but the same degree. These are designed for working professionals or candidates who miss the GATE-scholarship cutoff.

- IIT Hyderabad, IIT Jodhpur, IIT Gandhinagar
- NIT Surathkal (NIT Karnataka)

- Check individual institute portals for exact fee structures and eligibility criteria.

Chapter 5

Research & MS Programs

5.1 M.Tech Research vs. M.Tech (2-Year)

Factor	M.Tech TA (2 years)	M.Tech RA / Research (3 years min.)
Duration	2 years	3+ years
Stipend	₹12,400/month	₹13,400/month (approx.)
Course Load	Higher	Lower (more project-focused)
Research Output	Moderate	Higher (paper publications, PhD pathway)
Placements	Good	Same placement access at end of course
Guide	Optional	Must identify before joining

Pro-Tip: M.Tech Research Advantage

M.Tech Research is an excellent pathway to a **PhD from reputed institutes in India or abroad**. The research-focused environment enables high-quality project work and better paper publication opportunities.

5.2 Should I Pursue MS with a Low GATE Score?

Q: Should I pursue MS if I have a low GATE score?

A: Yes. Students with lower GATE scores still have MS admission opportunities, especially at newer IITs. While GATE score is the primary criterion, **interviews and**

undergraduate projects also significantly influence selection for MS programs. Prepare your projects well and aim for a strong interview performance.

5.3 Statement of Purpose (SOP)

Q: Does the MS program require an SOP?

A: Yes, most research-based MS programs (especially at IISc) require an SOP as part of the application. The SOP should clearly articulate:

- ✓ Your research interests and why you have chosen this area.
- ✓ Relevant UG projects, publications, or internships.
- ✓ Your long-term career goals.
- ✓ Why this specific institute and research group.

5.4 Scope of M.Tech Research in ECE at IISc

The **M.Tech Research** program at IISc is research-intensive with a reduced course load compared to the 2-year M.Tech. Key features:

- ✓ Work under a specific guide in a domain of your research interest.
- ✓ Excellent pathway for PhD admissions globally.
- ✓ IISc's strong publication culture means high-impact research output is achievable during the program.
- ✓ For **Photonic Integrated Circuits** research: IISc has dedicated labs; check the DESE department for relevant faculty.
- ✓ Placement opportunities exist at the end of the program for students who wish to enter industry.

Chapter 6

Drop Year & Career Guidance

6.1 Should I Take a Drop?

Q: Should I take a drop or join a self-sponsored NIT program?

A: If this was your **first attempt** and you have genuine interest in pursuing M.Tech from an IIT/IISc for the long-term career benefits:

- ✓ Taking a drop is a reasonable decision.
- ✓ Top branches at NITs like Trichy and Warangal also offer excellent placements if you don't wish to drop.
- ✓ Self-sponsored programs at top NITs are valid if you have no desire to re-attempt GATE.

Key question to ask yourself: *Do I genuinely want to be in the tech domain and pursue research/high-end engineering, or am I seeking a stable job?* Your answer determines whether a drop is worthwhile.

6.2 What If I Don't Qualify for GATE?

Pro-Tip: Stay Motivated

Not clearing GATE is not the end. Identify your mistakes in this attempt, rectify your weak areas, and adopt an organized and structured preparation strategy for the next attempt.

Immediate next steps:

- Analyze your GATE answer key and pinpoint subject-wise weaknesses.
- Join a structured study group or coaching if self-study hasn't worked.
- Consider attempting in the next year with focused preparation.

6.3 SSC JE Alongside M.Tech?

Eligibility Alert: Important Caution

Preparing for **SSC JE** or other government exams simultaneously with an M.Tech program is generally a **poor choice**. It divides your attention and compromises performance in both.

- ✓ If your primary interest is government sector jobs, SSC JE may be a direct and efficient path.
- ✓ Joining M.Tech purely as a backup while preparing for other exams wastes the resources and opportunities the M.Tech provides.
- ✓ Make a clear decision: M.Tech for tech growth **or** SSC JE for government jobs.

6.4 Other Options If Not IIT/NIT

Option	Notes
IIITs (Hyderabad, Delhi, etc.)	Strong tech culture; IIIT-H in particular has excellent placement records.
BITS Pilani (HD program)	Working professional program; no GATE needed; see Chapter 8.
Jadavpur University (JU)	Reputable institution; good for candidates in eastern India.
DTU Delhi	Decent option; own entrance exam; good for Delhi-based candidates.
MS from abroad (NUS / Germany)	Strong option for high-achieving students open to international education.

Chapter 7

Institute-Wise Program Profiles

Pro-Tip: How to Use This Chapter

Each section below lists the M.Tech/MS programs available, the mode of selection (GATE-only vs. Interview), and useful preparation links. Always verify on the official institute website as programs and selection criteria change each year.

7.1 IISc Bangalore

M.Tech Programs (with Interview)

All M.Tech and M.Tech Research programs at IISc require an interview.

- ✓ Microelectronics & VLSI (MVLSI)
- ✓ Electronic Systems Engineering (ESE)
- ✓ Computer Science & Engineering (CSE)
- ✓ Artificial Intelligence (AI)
- ✓ Signal Processing
- ✓ Computational Data Science (CDS)

Eligibility Alert: IISc Cutoffs

- ✓ MVLSI: 900+ GATE score (UR)
- ✓ ESE: 850+ GATE score (UR)

7.2 IIT Delhi

M.Tech Programs

- ✓ Integrated Electronics & Circuits (IEC) — *Interview required*
- ✓ RF Design & Technology
- ✓ Communications Engineering

- ✓ Computer Technology
- ✓ Control and Automation
- ✓ Power Electronics, Electrical Machines and Drives
- ✓ Power Systems
- ✓ VLSI Tool Design & Technology (VDTT)
- ✓ Opto-electronics & Optical Communication
- ✓ Instrument Technology / Applied Optics / Solid State Materials
- ✓ Telecom Technology & Management

MSR Programs (Written Test & Technical Interview)

- ✓ Integrated Electronics, Circuits, and Devices
- ✓ VLSI Tool Design & Technology (VDTT)

Official Link: `https://academics.iitd.ac.in/content/master-technology-mtech`

7.3 IIT Kanpur

M.Tech Programs

- ✓ Microelectronics & VLSI
- ✓ Signal Processing, Communications & Networks
- ✓ RF and Microwave
- ✓ Photonics
- ✓ Power Engineering
- ✓ Control & Automation

MSR Programs (Written Test & Technical Interview)

- ✓ Microelectronics & VLSI
- ✓ Signal Processing, Communications & Networks
- ✓ RF and Microwave / Photonics

Pro-Tip: IITK VLSI Observation

Historically, all General category seats in IIT Kanpur's VLSI M.Tech have been filled by IIT B.Tech graduates, leaving very limited seats for external GATE candidates. Verify the latest data before investing heavily in the application.

7.4 IIT Kharagpur

Electrical Engineering Department

- ✓ EE1: Machine Drives and Power Electronics
- ✓ EE2: Control System Engineering
- ✓ EE3: Power and Energy Systems
- ✓ EE4: Instrumentation and Integrated Electronics (*Interview required*)
- ✓ EE5: Signal Processing and Machine Learning

Electronics & Electrical Communication Engineering

- ✓ EC2: Microelectronics and VLSI Circuits
- ✓ EC3: RF and Microwave Engineering
- ✓ EC4: Communications and Signal Processing
- ✓ EC5: Vision and Intelligent Systems

Note: EE department requires interviews; EC department generally does not.

7.5 IIT Roorkee

ECE Department

- ✓ Microelectronics & VLSI
- ✓ Communication Systems
- ✓ R.F & Microwave Engineering

EE Department

- ✓ Electric Drives and Power Electronics
- ✓ Instrumentation & Signal Processing
- ✓ Power System Engineering
- ✓ Systems & Control

7.6 IIT Guwahati

Program	Details
Communication Engineering	EEE Dept
Power Engineering	EEE Dept
Microelectronics, Photonics & RF Engineering	EEE Dept
Signal Processing & Machine Learning	EEE Dept
VLSI and Nanoelectronics	EEE Dept
Systems, Control and Automation	EEE Dept

Official Link: https://iitg.ac.in/eee/course-details.php

7.7 IIT Hyderabad

- ✓ Micro Electronics and VLSI
- ✓ Communications and Signal Processing
- ✓ Power Electronics and Power Systems
- ✓ Systems and Control
- ✓ *Interdisciplinary:* Integrated Circuits and Microsystems Packaging (GATE: EE, EC, ME, IN, MS, PHY)

Pro-Tip: IITH among Emerging IITs

IIT Hyderabad is considered the **best among emerging IITs** in terms of placements, infrastructure, teaching quality, and research output.

7.8 Other IITs — Quick Reference

Institute	Key M.Tech Programs
IIT BHU	Microelectronics Eng., VLSI Architecture, Digital Systems, Communication, Microwave
IIT Mandi	VLSI, Communication & Signal Processing
IIT Ropar	Micro & VLSI, Comm. & Signal Processing, Power Engg.
IIT Gandhinagar	ICDT, Power Engineering & Control Systems
IIT Patna	VLSI and Embedded Systems, Communication System Engg.
IIT Indore	Communication & Signal Processing (CSP), VLSI Design & Nano Electronics (VDN)
IIT Tirupati	Microelectronics & VLSI
IIT Jammu / Bhubaneswar / Dhanbad	ECE / EE programs; check official portals.

7.9 COAP Admission Flowchart

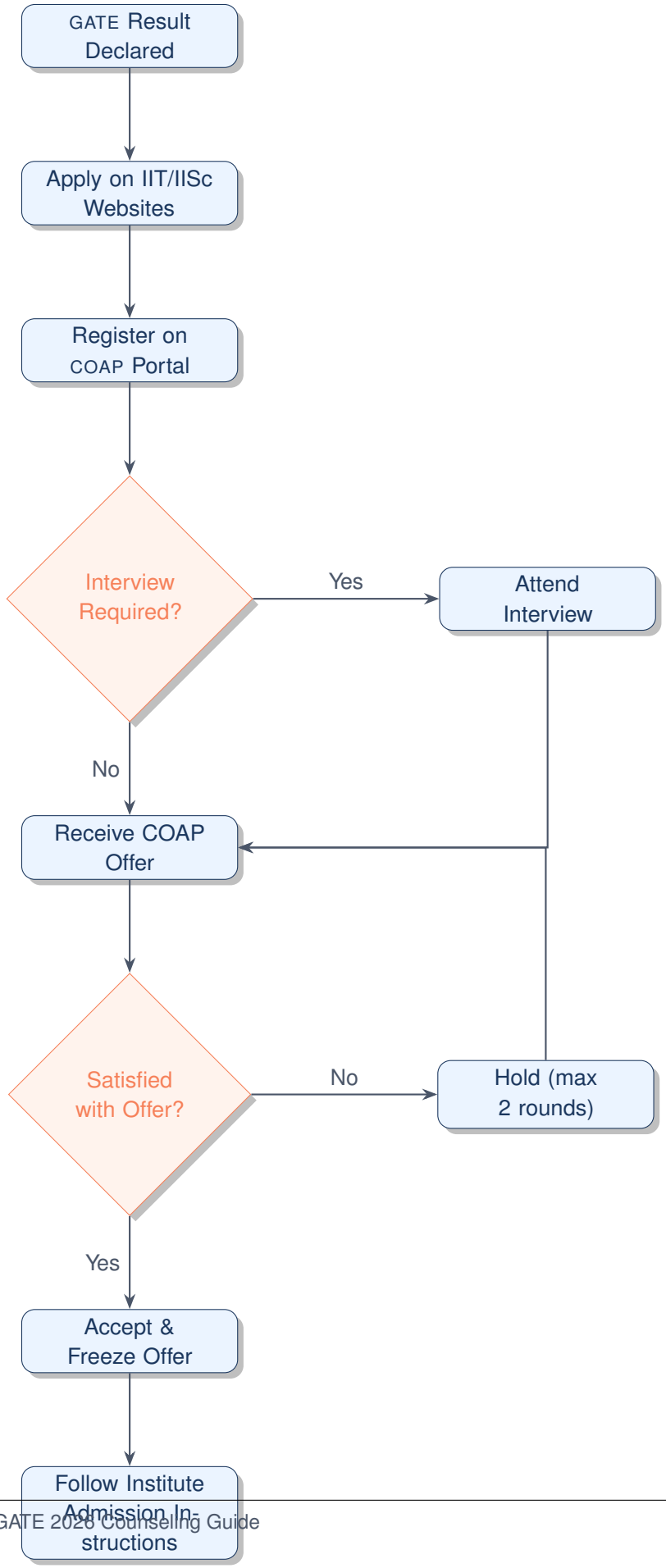

Figure 7.1. COAP Admission Process Flowchart

Chapter 8

Alternative Options for M.Tech

Pro-Tip: Second Chances Exist

These options provide a viable second chance but can be equally challenging to secure in their own right. Explore them seriously if GATE scores fall short of top IIT/NIT cutoffs.

8.1 BITS Pilani — Higher Degree (HD) Program

BITS Pilani's Higher Degree program is a work-integrated M.Tech for **working professionals and fresh graduates**.

Campus	VLSI-Related Branches
BITS Hyderabad	MVLSI, Embedded Systems
BITS Pilani	MVLSI, Embedded Systems
BITS Goa	MVLSI, Embedded Systems

Important Date: BITS HD 2025 Key Dates

Event	Date
Application Deadline	19th April 2025 by 5 PM
Online Test Dates	26th May & 1st June 2025

Apply: `https://pdc.edvertek.com/bitshdapply25pd/Default`

8.2 IIIT Hyderabad — PGEE

IIIT Hyderabad conducts its own **PG Entrance Examination** (**PGEE**) for:

- ✓ M.Tech
- ✓ Master of Science by Research
- ✓ Ph.D.

- PGEE Portal: https://pgadmissions.iiit.ac.in/pgee/
- M.Tech Fee Details: https://pgadmissions.iiit.ac.in/m-tech-program/
- MS/PhD Admissions: https://pgadmissions.iiit.ac.in/ms-phd-programs/

8.3 IIIT Delhi

- M.Tech Admissions: https://iiitd.ac.in/admission/mtech/2025
- M.Tech ECE Details 2025: https://iiitd.ac.in/admission/mtech/2025/ece-details

8.4 Other Alternatives

Institution	Notes
Indian Statistical Institute (ISI)	Admission via GATE (high cutoff) or separate written test + interview.
VIT Vellore (VITMEE)	Own entrance exam (VITMEE 2025 date: 20 April 2025). Good option for students who miss the GATE cutoff.
DA-IICT Gandhinagar	Separate entrance exam; no IIT tag but good placements; strong CS and ECE programs.
NUS / Germany (MS abroad)	Excellent option for high achievers considering global education; no GATE needed; requires GRE/IELTS in some cases.
Jadavpur University / DTU	Comparable to mid-tier NITs; own entrance process.

8.5 Masters in Power Electronics from IITs

Pro-Tip: External Resource

A detailed FAQ document on Masters in Power Electronics from IITs has been compiled separately by the PrepFusion community. Access it at: https://docs.google.com/document/d/1yl0VyGRHbqXeNTREY2ew-Q373Jo6a-8nSs2K6hLbJBw/edit?usp=sharing

Part II

PSU Recruitment

Chapter 9

PSU Recruitment & Preparation

Pro-Tip: Chapter Overview

Public Sector Undertakings (PSUs) offer stable, high-paying careers with strong growth prospects. This chapter covers the recruitment process, cutoffs, and strategic preparation tips sourced directly from PrepFusion students.

9.1 Overview of PSU Recruitment Modes

PSU recruitment has evolved significantly in recent years. Two primary modes now co-exist:

Mode	Description	Examples
GATE Score Based	Recruitment purely through valid GATE rank, followed by GD & interview.	IOCL, GAIL, NTPC, BHEL, BEL, PGCIL
Own Exam Based	PSU conducts its own written examination followed by GD & interview.	DRDO, some state PSUs

Eligibility Alert: GATE Score Validity for PSUs

- ✓ **Most PSUs:** Accept current year GATE rank only.
- ✓ **BARC:** Accepts GATE scores up to **2 years** old.
- ✓ **NPCIL:** Accepts GATE scores up to **3 years** old (final selection based entirely on technical interview).

9.2 Recruitment Process (Step-by-Step)

Stage 1: Written Test / GATE Shortlisting. Candidates are shortlisted based on their GATE rank or the PSU's own written exam score. For GATE-based PSUs, typically only the **current year** rank is valid.

Stage 2: **Group Discussion (GD).** Many PSUs include a GD round to evaluate communication skills, situational awareness, and team dynamics.

Stage 3: **Technical Interview.** Subjects tested vary by discipline. For EE/ECE:

- ✓ Power Systems (protection, stability)
- ✓ Electrical Machines (synchronous, induction, transformers)
- ✓ Analog & Digital Circuits
- ✓ Company-specific topics (business, chairman/MD name, projects)

Stage 4: **HR / Personal Interview.** Expect questions on: Why PSU over private sector? Where do you see yourself in 10 years? Why this specific PSU?

Stage 5: **Medical Examination.** A rigorous medical clearance is mandatory before final selection. Multiple tests must be cleared; failure at this stage results in disqualification.

9.3 Score Weightage Breakdown

PSU	GATE Score	GD	Interview
IOCL	80–85%	15–20% combined	
GAIL	80–85%	15–20% combined	
NTPC	Rank-based shortlist	Interview-heavy	
BARC	Qualifying only	100% technical interview	
NPCIL	Qualifying only	100% technical interview	

9.4 Rank-wise PSU Expectation (ECE, OBC Category)

GATE Rank Range	Targetable PSUs	Notes
Under 150 (General)	Top PSUs (IOCL, GAIL, BHEL, PGCIL, NTPC)	Safe zone for general category in most PSUs.
150–500	NTPC, BEL, BHEL, CIL, NPCIL	Apply broadly; selection depends on GD/interview.
1300–1400 (OBC)	NTPC (Round 2, cutoff ≈1118), CIL (Round 3)	NPCIL 2024 OBC cutoff was ≈563 GATE score.
40 marks (≈560 score)	Selected NITs via CCMT; BITS Pilani (HD program); PhD interviews at IITs	Visit CCMT website for NIT VLSI branch cutoffs.

9.5 Career Growth in PSUs

Years of Service	Typical Designation
0–5 years	Executive / Junior Manager
5–15 years	Senior Manager / AGM
15–25 years	DGM / GM
25–30+ years	ED / Director / Chairman (rare, high-performers)

Pro-Tip: PSU Career Advantages

Beyond promotions, many PSUs:

- ✓ Assign responsibilities *outside* your engineering domain, broadening your skill set.
- ✓ Sponsor training sessions run by top technology companies, yielding industry-recognized certifications.
- ✓ Offer excellent work-life balance, medical benefits, and housing allowances.

9.6 PSU Interview Preparation Strategy

9.6.1 Technical Preparation

- ✓ **Power Systems:** Protection schemes, stability analysis (for EE).

- ✓ **Electrical Machines:** Synchronous machines, induction motors, transformers—ratings, nameplate data, working principles.
- ✓ **Analog & Digital:** Core ECE topics; expect basic definition questions (e.g., "Define a Flip-Flop").
- ✓ **Final Year Project:** Especially for PSUs like **Powergrid** and **EIL**, which deep-dive into your project. Prepare your BIODATA accordingly.
- ✓ **BEL/BHEL Strategy:** Practice previous-year questions (easier than GATE). Solve in a time-bound manner. Focus on practical aspects and theory.

9.6.2 Non-Technical / HR Preparation

- ✓ Research the company: Chairman, MD, major projects, financial figures.
- ✓ Prepare answers for: "Why PSU over a private company?"
- ✓ Prepare answers for: "Why not M.Tech if you are a fresher?"
- ✓ Research papers and internships are **less critical** for PSU interviews, except for EIL where practical experience matters.

9.6.3 M.Tech vs. Direct PSU

Eligibility Alert: M.Tech On-Campus Placement Rule

M.Tech on-campus placement is based **entirely on M.Tech CGPA**. Your B.Tech CGPA has **no role** in campus recruitment during M.Tech. This is a significant advantage for candidates with a lower B.Tech GPA.

9.7 Frequently Asked Questions

Q: What is the weightage of research papers and final year projects in PSU interviews?

A: Research papers are generally **not important** for most PSU interviews. However, practical industry experience, internships, and final year projects play a significant role in some PSUs like **EIL**. For Powergrid, interviewers deep-dive into the projects mentioned on your BIODATA form.

Q: What are the opportunities for a candidate scoring 40 marks in GATE ECE 2025?

A: In GATE 2024, 40 marks corresponded to approximately a 560 score. At this level, PhD interview opportunities exist at IITs/IISc (interview cutoffs are very low). CCMT provides access to some NIT VLSI-related branches. BITS Pilani HD programs are also a good option if fees are not a constraint.

Chapter 10

Interview Preparation

10.1 General Strategy

Pro-Tip: Core Principle

GATE preparation covers the majority of the technical syllabus. Shift your focus from *formula memorization* to *conceptual understanding*—IISc panels in particular emphasize fundamentals over rote recall.

10.1.1 For M.Tech Interviews (IITs)

- ✓ Master EC core subjects: **Analog** and **Digital circuits**.
- ✓ Do not neglect **Communications**—IIT panels ask from it occasionally.
- ✓ For AI/ML branches: basic **C coding**, **linear algebra**, and **matrices** are frequently tested.
- ✓ Focus on **fundamentals**, not just formulae.

10.1.2 For M.Tech Research / MS RA Interviews

- ✓ Prepare your UG final year project and any published papers thoroughly. Interviews often involve a discussion of your own work.
- ✓ Expect follow-up questions on the **technology behind your project**.
- ✓ IISc panels are particularly rigorous—expect whiteboard problem-solving.

10.1.3 For PSU Interviews

See Chapter 9, Section 9.6 for a detailed strategy.

10.2 Branch-wise Focus Areas

Branch / Domain	Key Topics to Prepare
VLSI / Microelectronics (ECE)	Analog circuits, Digital circuits, Device physics, CMOS design, Transistor-level design
Signal Processing / Communications	DSP fundamentals, Modulation, Fourier analysis, Filter design
Power Electronics / EE (IITs)	Power system protection, Stability analysis, Electrical machines, Power electronics
AI/ML (Interdisciplinary)	Linear algebra, Probability, Basic C/Python, Machine learning fundamentals
IISc (all branches)	Heavy emphasis on fundamentals; expect whiteboard derivations and follow-up questions on incorrect answers

10.3 Recommended Preparation Resources

Subject	Resource	Link / Notes
All Subjects (Interview)	PrepFusion Free Course	https://prepfusion.in/new-courses/46/content
Electrical Networks & RC Circuits	NPTEL – Dr. Nagendra Krishnapura	Available on YouTube / NPTEL website
Analog Circuits (BJT, MOS, Diode)	NPTEL – Dr. Nagendra Krishnapura	Same playlist as Networks
Digital Circuits & Math	GATE Syllabus + Standard textbooks	Sufficient for most M.Tech interviews
IISc Interview Experiences	PrepFusion Interview Prep Video	https://youtu.be/gjFeuFKBALQ

Pro-Tip: Free vs. Paid Courses

PrepFusion's free courses cover all fundamental concepts required for M.Tech interviews. Premium options offer additional technical and aptitude-based training. The free course is an excellent starting point before investing in paid

resources.

10.4 Preparing for the IAP Branch

For the **Integrated Analog Pathfinder (IAP)** branch or similar analog-focused programs at IISc:

- ✓ Study **oscillator design** thoroughly (Wien Bridge, Colpitts, etc.).
- ✓ Understand **negative feedback** deeply—its effect on gain, bandwidth, and stability.
- ✓ Practice **op-amp circuit** analysis on paper/whiteboard.
- ✓ Revise **differential amplifiers**: small-signal model, CMRR, differential gain.
- ✓ Brush up on **control systems** fundamentals (dominant pole, pole-zero plots, step response).

10.5 Frequently Asked Questions

Q: What kind of questions are asked in interviews for VLSI and related branches?

A: It depends on the panel. Generally, questions come from **analog** and **digital circuits**. Device physics questions may also be asked to EC students. IISc panels additionally ask **power electronics** questions of EE students and occasionally **communications** questions of EC students.

Q: How to prepare for MS and M.Tech RA program interviews?

A: Similar to M.Tech interviews, but with an additional component: the **projects and papers** you mentioned in your application are actively discussed. Prepare to defend your research interests and explain technical choices you made during your UG project.

Appendix A

Technical Interview Question Bank

Pro-Tip: How to Use This Appendix

Questions are organized by domain. Practice solving them on paper—many IIT and IISc interviews require whiteboard solutions. For digital questions, also draw timing diagrams and logic circuits where applicable.

A.1 Digital Circuits

A.1.1 Number Systems & Representation

D1. Convert the decimal number 156.75 into its binary, octal, and hexadecimal equivalents.

D2. Convert the following decimal numbers into signed binary, octal, and hexadecimal using the minimum possible number of bits:

(a) $+17$

(b) -17

D3. Explain the difference between 1's complement and 2's complement representations of a negative number. Why is 2's complement more widely used in computer systems?

D4. Explain the 2's complement representation of negative numbers. Convert 13 and -13 into:

(a) 8-bit signed magnitude

(b) 8-bit 1's complement

(c) 8-bit 2's complement

D5. What is the difference between binary and Gray code? Why is Gray code used in applications like rotary encoders?

A.1.2 Combinational & Sequential Logic

D6. Define a Flip-Flop. What is the difference between a latch and a flip-flop?

D7. Design a 4-bit binary multiplier circuit using basic gates. Draw the schematic on the whiteboard.

D8. Explain how a Wallace Tree multiplier works. How does it improve speed over a simple array multiplier?

D9. Sketch the timing diagram for a 2-bit multiplier with inputs $A = 11_2$ and $B = 10_2$. Show intermediate and final outputs.

A.2 Analog Circuits

A.2.1 Oscillators & Feedback

A1. Draw the circuit of a Wien Bridge oscillator and explain how it sustains oscillations. What sets its oscillation frequency?

A2. Sketch the output waveform of an oscillator you designed. How would you stabilize its amplitude?

A3. Explain the role of negative feedback in an amplifier circuit. How does it affect gain and stability?

A4. Design an op-amp circuit with negative feedback on the whiteboard. Derive its closed-loop gain.

A5. What happens to an oscillator if positive feedback becomes too strong? Illustrate with a block diagram.

A.2.2 Control Systems (Analog Context)

A6. What is the dominant pole concept in control systems? How does it influence system response?

A7. Sketch a second-order system's pole-zero plot and explain its step response.

A8. How would you determine the stability of a feedback system using pole locations? Show your approach.

A.2.3 Op-Amp Circuits

A9. Given an op-amp circuit [provided by interviewer]: solve for the output voltage on the whiteboard and explain each step.

A10. Modify a given op-amp circuit to act as an integrator. Draw the new schematic and its output waveform for a square-wave input.

A.2.4 Differential Amplifiers

A11. Draw a differential amplifier using MOSFETs. Calculate its differential gain and show the small-signal model.

A12. What happens to the CMRR if there is a mismatch in the differential pair? Explain with a diagram.

A13. Sketch the input and output waveforms of a differential amplifier for a sinusoidal differential input.

A.3 Power Systems (EE Domain)

P1. Explain the working principle of a **synchronous machine** and describe the role of the excitation system.

P2. What is the difference between an **induction motor** and a synchronous motor? Describe starting methods for induction motors.

P3. Explain **power system protection** principles: What is a differential relay and when is it used?

P4. Define **transient stability** in a power system. Sketch the equal area criterion for a fault.

P5. What are the nameplates and typical ratings of a 100 kVA, 11 kV/415 V distribution transformer?

A.4 IISc Analog Domain Interview (Sample Session)

Interview Experience | Anonymous Candidate (*IISc Bangalore – Analog Domain*)

Interviewer: Prof. Aniruddhan **Duration:** 30–35 minutes

Focus areas: Oscillator design, Feedback concepts, Control Systems, Op-Amp circuits, Differential amplifiers.

Style: Intense, whiteboard problem-solving, follow-ups on written test mistakes.

- ✓ Draw a Wien Bridge oscillator and explain oscillation sustainment.
- ✓ Design an op-amp circuit with negative feedback; derive closed-loop gain on the whiteboard.
- ✓ Explain the dominant pole concept and sketch a second-order system's step response.
- ✓ Draw a MOSFET differential amplifier; calculate differential gain and show the small-signal model.

- ✓ What happens to CMRR with mismatch in the differential pair?

Tip: Interviewers at IISc often revisit questions you answered incorrectly in the written test. Prepare to solve similar problems on the spot.

A.5 IISc Digital Domain Interview (Sample Session)

Interview Experience | Anonymous Candidate (*IISc Bangalore – Digital Domain*)

Interviewers: Prof. Janakiraman & Prof. Vinita Vasudevan **Duration:** ≈10 minutes

Focus: Multiplier circuits, domain interest.

- ✓ Design a 4-bit binary multiplier using basic gates.
- ✓ Explain Wallace Tree multiplier and its speed advantage.
- ✓ Draw the timing diagram for a 2-bit multiplier: $A = 11_2$, $B = 10_2$.

Note: The panel also asked about the candidate's genuine interest in the digital domain vs. analog, and requested an example of a digital circuit they had worked on.

Appendix B

Student Interview Experiences

Disclaimer

All interview experiences below are shared voluntarily by **PrepFusion** students. Question sets, panel compositions, and processes **vary each year** and may differ from what is documented here. Use these accounts for orientation only—always verify current details with the respective institute.

B.1 How to Contribute Your Experience

If you have recently appeared for an IIT or IISc interview, please contribute your experience to help future students! Share via the PrepFusion Google Doc:

`https://docs.google.com/document/d/1WaXfhvyS9vxOzVhDEFRc2AwDmobg-vkDxYCtel89IqU`

Our interns will format your contribution using the `template_interview.tex` file and include it in the next update of this guide.

B.2 Experience Template Reference

Interview Experience | [Student Name or Anonymous] (*[Institute] – [Branch/Program]*)

Date: [Month, Year] **Duration:** [Duration of interview]

Written Test: [Describe the written test format, difficulty, and topics covered.]

Interview Panel: [Professor names / Anonymous]

Questions Asked:

Question 1

Question 2

r remove as needed

Key Takeaways / Tips for Future Candidates:

Tip 1

Tip 2

B.3 IISc Bangalore

Interview Experience | Contributed via PrepFusion Community (*IISc Bangalore – MVLSI*)

Written Test: Standard analog + digital MCQ format similar to GATE difficulty. Some questions on control systems.

Analog Interview (Prof. Aniruddhan):

- ✓ Wien Bridge oscillator — draw, explain, what sets frequency?
- ✓ Op-amp with negative feedback — draw, derive closed-loop gain.
- ✓ Dominant pole concept in control systems.
- ✓ MOSFET differential amplifier — small-signal model and CMRR.
- ✓ Written test correction on whiteboard.

Digital Interview (Prof. Janakiraman & Prof. V. Vasudevan):

- ✓ Design a 4-bit binary multiplier using basic gates.
- ✓ Explain Wallace Tree multiplier.
- ✓ Domain interest check: Are you genuinely interested in digital or analog?

Key Tip: IISc emphasizes **fundamentals** over rote formulas. Expect follow-up questions on anything you answered incorrectly in the written test. Whiteboard confidence is essential.

B.4 IIT Delhi

Interview Experience | [Placeholder] (*IIT Delhi – IEC*)

This section will be filled with community-sourced interview experiences. If you appeared for an IIT Delhi IEC interview, please share your experience at the link above.

B.5 IIT Bombay

Interview Experience | [Placeholder] (*IIT Bombay – ICS / ESE*)

This section will be filled with community-sourced interview experiences. If you appeared for an IIT Bombay MS or M.Tech Research interview, please share your experience.